AirCast: Predicting Urban PM2.5 Concentration with Classical Regression

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Summary

This report builds and evaluates regression models that estimate hourly PM2.5 concentration at twelve air-quality monitoring stations in Beijing from meteorological and calendar features. The dataset contains 420,768 hourly observations from March 2013 to February 2017. It was split chronologically into a two-year training set, a one-year validation set and a one-year test set, and all preprocessing was learned from the training data only. Co-pollutants such as PM10 and CO were excluded from the inputs, because they are measured at the same moment as PM2.5 and would inflate accuracy without reflecting genuine predictive ability.

Five regression models were compared against a mean-only baseline. A tuned Random Forest gave the best validation performance (RMSE 64.7 μg/m³, R^2 0.375), compared with 67.3 μg/m³ and 0.325 for the Linear Regression baseline. On the unseen test year it achieved □FILL: MAE, RMSE, R^2□. The models explain roughly a third of the variation in PM2.5. They are reliable for moderate pollution levels but systematically underpredict severe winter episodes, which is the main limitation identified.

1. Dataset

The data is the Beijing Multi-Site Air-Quality dataset, available from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data) and mirrored on Kaggle. It records hourly measurements at 12 stations from 1 March 2013 to 28 February 2017. Each station file has 35,064 rows, exactly 1,461 days × 24 hours, so the time index is complete. Missing data appears only as missing values within rows, never as missing hours.

Data quality. At the Aotizhongxin station, missing rates were 2.6% for PM2.5, 5.1% for CO and 4.9% for O3, while every meteorological variable was below 0.1%. There were no negative or zero PM2.5 readings and no negative wind speeds. The checks found □FILL□ fully duplicated rows and □FILL□ duplicated station-hour pairs. They also found □FILL□ hours where PM2.5 exceeded PM10. That is physically implausible, because PM10 includes all PM2.5 particles, so those hours point to instrument error. Several early readings show PM2.5 exactly equal to PM10 at very low concentrations, which may reflect instrument limits. CO is reported on a much larger numerical scale than the other pollutants, but it was not used as a model input.

Limitations of the data. It covers one city over a four-year period that predates later pollution-control policies, so results may not generalise elsewhere. Wind direction is coarsely categorised. Station readings describe ambient concentration at a fixed point, not personal exposure.

2. Exploratory Data Analysis

Detailed exploration used a single central station, Aotizhongxin. Cross-station patterns used all twelve stations, which is also the data the models are trained on.

2.1 Distribution of PM2.5

PM2.5 is strongly right-skewed (Figures 1 and 2). The mean (82.8 μg/m³) is well above the median (58 μg/m³), and the standard deviation (82.1) is about as large as the mean. Half of all hours fall between 22 and 114 μg/m³, but the tail extends to 898 μg/m³, with 76 hours above 500.

This shape has two consequences for the analysis. Squared-error metrics such as RMSE and R^2 are dominated by the rare extreme hours. Methods that assume normality, such as the z-score rule, are unreliable here.

2.2 Temporal and location patterns

The daily-average series (Figure 3) shows sharp, short-lived pollution episodes rather than a smooth trend. As Section 3 confirms, these episodes cluster heavily in the cold months.

Across stations (Figure 6), mean PM2.5 ranges from 66.0 μg/m³ at Dingling to 86.2 μg/m³ at Dongsi. The suburban and northern stations (Dingling, Huairou, Changping) are the cleanest, and the urban-core stations (Dongsi, Wanshouxigong, Nongzhanguan, Gucheng) are the most polluted. Monthly means peak in winter. By hour of day, □FILL: describe when the hourly curve peaks and dips in your pm25_patterns plot□.

2.3 Relationships with explanatory variables

The correlation heatmap (Figure 4) shows that PM2.5 is most strongly correlated with the other pollutants.This is why they were excluded as inputs. They move with PM2.5 because they share the same sources and the same weather conditions, so using them would amount to measuring pollution rather than predicting it.

Among the meteorological variables, □FILL: the strongest relationships, e.g. wind speed r = __, dew point r = __□. The scatter plots (Figure 5) show □FILL: confirm against your plot, e.g. whether the highest concentrations occur mainly at low wind speeds□. That pattern would be consistent with stagnant air allowing pollutants to accumulate. These are associations, not causal effects.

3. Outlier Detection

Two methods were applied to PM2.5 at Aotizhongxin.

Every hour flagged by the z-score rule was also flagged by the IQR rule, and 1,033 hours were flagged by IQR alone. The two methods disagree because of the skew. The IQR lower fence is negative (−116), so it can only flag high values. The z-score threshold (mean + 3 standard deviations) lies above the IQR fence because the long tail inflates the standard deviation. The z-score method assumes a roughly normal distribution, which PM2.5 clearly is not, so the IQR result is the more trustworthy of the two. The same pattern held for the other variables: CO (2,607 IQR vs 741 z-score

flags), wind speed (1,742 vs 500), PM10 (1,080 vs 508) and NO2 (472 vs 326). Temperature had no outliers under either rule.

Decision: all observations were retained. Of the 1,624 IQR-flagged hours, 1,506 (about 93%) fall between October and March, the period of coal-based winter heating in Beijing. A sensor fault would not cluster seasonally like this, so these hours are best interpreted as genuine severe-pollution events. Removing them would make the models look better while making them useless in exactly the situations where accurate estimates matter most.

4. Preprocessing and Feature Engineering

Target and inputs. The target is hourly PM2.5. The inputs are the five meteorological variables (TEMP, PRES, DEWP, RAIN, WSPM), wind direction, station, and the engineered calendar features below. That gives 39 inputs after one-hot encoding.

All engineered features come from the timestamp of the observation itself. None of them uses any future information.

Missing values and encoding. Rows with a missing target were dropped rather than imputed, because imputing the target would invent training labels. Missing numeric inputs were filled with the training-set median, and missing wind directions with the training-set mode. Since meteorological missingness was below 0.1%, this choice has negligible effect. Numeric features were standardised. Wind direction and station were one-hot encoded, with unseen categories ignored.

Leakage control. The split is chronological: training covers March 2013 to February 2015 (205,989 hours with a valid target), validation March 2015 to February 2016 (103,287 hours) and test March 2016 to February 2017 (102,753 hours). Each period covers complete years, so every season appears in every set. The imputation, scaling and encoding steps sit inside a scikit-learn `Pipeline`, so they are fitted on training data only and simply applied to the validation and test sets. The test set was used once, for final reporting.

5. Models and Validation Strategy

Linear Regression served as the main baseline, alongside a mean-only predictor that always outputs the training average and so sets the floor any useful model must beat. The other models were Ridge, Lasso, a Decision Tree (depth 8, at least 50 samples per leaf) and a Random Forest. Every model uses a random seed of 42 for reproducibility.

Random Forest tuning. The first Random Forest (depth 15, at least 20 samples per leaf) overfit heavily, with a train R^2 of 0.779 against a validation R^2 of 0.310. Three configurations were therefore compared on the validation set.

The middle setting gave the lowest validation RMSE and the best R^2, and it narrowed the train–validation gap considerably. It was selected and used with 50 trees and half the features considered at each split.

Regularisation (D7). Ridge with $\alpha = 1$ produced results identical to Linear Regression. An α grid gave □FILL: validation RMSE for α = 1, 10, 100, 1,000, 10,000□. Lasso ($\alpha = 0.1$) retained □FILL□ of 39 coefficients and matched Linear Regression almost exactly (validation RMSE 67.29, R^2 0.325). Regularisation adds little here. With about 206,000 training rows and only 39 inputs, the unregularised model is already stable and not overfitting: its train and validation R^2 are 0.386 and 0.325. Ridge and Lasso help most when features are many relative to observations, which is not the case in this problem.

Time-series cross-validation (D8). Standard k-fold cross-validation shuffles rows, letting the model train on future hours and test on past ones. Instead, `TimeSeriesSplit` with four expanding folds was applied to the combined training and validation periods. Each fold trains on all earlier data and tests on the block that follows.

The Random Forest fails in fold 1 but beats Linear Regression in folds 3 and 4. Fold 1 trains on roughly the first seven months of data, spring and summer 2013, and is then tested on the following winter. The forest has never seen a heating season, and because trees cannot predict beyond the range of their training targets, it cannot produce winter-level concentrations. Linear Regression extrapolates through its seasonal and dew-point terms, so it degrades more gracefully.

Once training data includes at least one winter, the Random Forest is the stronger model. Its lower mean MAE across folds also shows it is more accurate on typical hours. The fold 1 result is a useful warning: tree models need training data that covers the full range of conditions they will face.

6. Results

The metrics measure different aspects of error.

- **MAE** is the average absolute error in μg/m³, and it is the easiest to interpret.
- **MSE** averages squared errors, so it penalises large mistakes heavily.
- **RMSE** is the square root of MSE, which puts it back in μg/m³ while keeping the emphasis on large errors.
- **R²** is the share of variance in PM2.5 explained, relative to always predicting the mean.

Selected model: the tuned Random Forest. It achieved the lowest validation RMSE and MAE and the highest validation R² of all candidates. It also cannot output negative concentrations, whereas Linear Regression predicted negative values for 6.3% of test hours. All models clearly beat the mean-only baseline, reducing test MAE from 61 μg/m³ to the mid-40s.

The improvement of the tuned forest over Linear Regression is real but modest, about 2.6 μg/m³ in validation RMSE and 0.05 in R². It is practically meaningful mainly in how it handles typical hours, not as a step change in accuracy.

Test R² for the linear models (0.374) is higher than their validation R² (0.325). This reflects year-to-year variation in pollution rather than a flaw in the method. The validation year includes the severe episodes of December 2015, when Beijing issued its first red pollution alerts, and those extreme hours are inherently harder to estimate.

7. Error Analysis and Interpretation

Errors by pollution range. Residuals are defined as actual minus predicted, so positive values mean the model underpredicted.

The errors are systematic, not random. The baseline model overpredicts clean hours by about 30 μg/m³ and underpredicts the most polluted hours by about 186 μg/m³ on average, which is classic regression toward the mean. It is most accurate in the 75–150 range, near the centre of the training distribution.

The residual plot shows a funnel shape: residual spread grows as predictions increase. The variance of the errors is therefore not constant. □FILL: one sentence on whether the tuned forest shows the same pattern, and how much it reduces the error at 250+□.

Errors by time and location. The baseline model's monthly MAE ranges from about 33 μg/m³ in June to 73.6 in January, and December (70.1) is similarly poor. Accuracy roughly halves in winter, matching the seasonal clustering of outliers seen in Section 3. Across stations, MAE is lowest at the cleaner suburban sites (Dingling 44.2, Huairou 45.1) and highest in the urban core (Dongsi 50.3, Wanshouxigong 49.3), so the models struggle more where pollution is higher and more variable.

Influential predictors. Permutation importance measures how much validation R^2 falls when one feature's values are randomly shuffled (Figure 9). For the tuned forest, the most influential features were □FILL: top three□. In the Linear Regression baseline, dew point, the seasonal terms and temperature dominated.

These results should be read with two cautions. First, correlated features such as dew point, temperature and month inflate each other's importance, because shuffling one breaks combinations the model relies on. Second, importance reflects association, not causation. Dew point matters because humid, stagnant winter conditions coincide with pollution build-up, not because moisture itself creates particulate matter.

8. Limitations and Improvements

No information about recent pollution. The models see only current weather and the calendar, never past PM2.5. Pollution episodes build up over hours to days, so this is the main reason R^2 stays below 0.4. Lagged PM2.5 values and rolling averages, built

strictly from earlier hours and evaluated with the same chronological split, would likely give the largest single improvement.

Concurrent rather than forecast weather. The models use weather measured in the same hour, which makes this an estimation task rather than a true forecast. In a deployed forecasting system, those inputs would have to come from weather forecasts, which carry their own errors.

Skewed target and systematic underprediction of extremes. Severe episodes are both rare and the most important to get right. Modelling log(PM2.5), weighting extreme hours more heavily during training, or using gradient-boosted trees could reduce the large errors above 250 μg/m³.

Limited coverage. The data comes from one city and one four-year period. Applying the model to another city, or to Beijing after later emission policies, would require retraining and fresh validation.

9. Conclusion

Using only meteorological and calendar information, classical regression models explain roughly a third of the hourly variation in Beijing's PM2.5. They substantially outperform a naive baseline. A tuned Random Forest performed best on validation, while time-series cross-validation showed that tree models need training data that spans every season before they can be trusted.

The strongest associations with PM2.5 were humidity-related weather conditions and seasonality, consistent with winter heating episodes. The main weakness is the systematic underestimation of severe pollution, which is exactly where accurate estimates matter most. Adding recent pollution history under chronological validation is the clearest next step.